

Scientific Communities in the Literature Selection

Qualitative coding report based on title and keyword fields

This report inductively groups the uploaded literature selection into thematic scientific communities. The coding uses only the title and keywords columns of the CSV file. The dataset contains 250 records. Categories are intentionally interpretive and not mutually exclusive: one paper may contribute to several communities when its title or keywords point to multiple fields.

Coding approach

I first scanned recurring keywords and title phrases, then consolidated overlapping terms into higher-level research communities. Counts are indicative multi-label frequencies, not exclusive cluster sizes. The categories below therefore describe recognizable communities represented in the literature selection rather than a strict classification with only one code per article.

Overview of coded communities

Community category	Indicative count	Definition	Example from the file
Higher Education, College Access and Student Success	54	Studies centered on postsecondary settings, college access, undergraduate/graduate trajectories, persistence, advising, transitions, and student success in universities or colleges.	Going to College With a Posse: How Having High School Peers on Campus Supports College Achievement Keywords: admissions; equity; first-generation college students; hierarchical linear modeling; higher education; peer interaction/friendship; privilege; psychology; regression analyses; retention; social belonging; social capital; socioeconomic status
K-12 Schooling, Curriculum and Educational Policy	29	Work addressing compulsory schooling, curriculum, textbooks, school policy, school districts, absenteeism, and primary/secondary educational reform.	Academic Harms of Missing High School and the Accuracy of Current Policy Thresholds: Analysis of Preregistered Administrative Data From a California School District Keywords: absenteeism; high school; preregistration; secondary data
STEM, Engineering and Disciplinary Science Education	55	Research communities focused on science, technology, engineering, mathematics, computing, life sciences, bioinformatics, hydrology, energy, or other disciplinary learning contexts.	Revisiting barriers to implementation of bioinformatics into life sciences education Keywords: undergraduate education; STEM education; bioinformatics; barriers to integration; education reform
Artificial Intelligence, Learning Analytics and Educational Data Science	48	Studies using or examining AI, machine learning, NLP, learning analytics, automated scoring, predictive models, LLMs, topic modeling, or educational data science.	Explainable Automated Essay Scoring: Deep Learning Really Has Pedagogical Value Keywords: explainable artificial intelligence; SHAP; automated essay scoring; deep learning; trust; learning analytics; feedback; rubric
Digital, Online, Immersive and Game-Based Learning	55	Communities around technology-enhanced learning environments, online/remote learning, educational technology, AR/VR, games, simulations, tutoring systems, multimedia, and digital tools.	The more the better? Comparing two SQD-based learning designs in a teacher training on augmented and virtual reality Keywords: WillSkillTool model; Teacher education; Teacher training; SQD model; Augmented reality; Virtual reality; Learning design; Learning events
Teacher Education, Professional Learning and Teaching Practice	51	Research on teachers, instructors, teacher educators, professional development, teacher training, pedagogical practice, teaching quality, and faculty-student instructional relationships.	Bolstering spatial learning in the primary classroom: identifying the factors underlying primary teachers spatial pedagogical practices Keywords: teacher cognition; spatial skills; spatial anxiety; spatial habits of mind; primary education
Equity, Inclusion, Diversity and Special Education	45	Studies foregrounding educational inequality, inclusion, diversity, race/ethnicity, gender, socioeconomic disadvantage, gifted education, special education, disability, access, and widening participation.	Effect of Local Norms on Racial and Ethnic Representation in Gifted Education Keywords: assessment; descriptive analysis; diversity; equity; gifted education; hierarchical linear modeling; policy analysis

Community category	Indicative count	Definition	Example from the file
Wellbeing, Mental Health and Social-Emotional Development	18	Research focused on psychological wellbeing, mental health, stress, emotion, belonging, social-emotional outcomes, resilience, and school or teacher wellbeing.	Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources Keywords: COVID-19 pandemic, college students, psychological well-being, emotions, experience sampling method, personality, intraindividual trajectories
Assessment, Psychometrics and Measurement	72	Measurement-oriented work on assessment, tests, scales, validation, psychometrics, Rasch/SEM models, automated or human evaluation, screening tools, and measurement quality.	A Model-Based Examination of Scale Effects in Student Evaluations of Teaching Keywords: student evaluations of teaching; response scales; scale mapping
Motivation, Engagement, Identity and Self-Regulated Learning	49	Communities studying learners attitudes, identity, motivation, self-efficacy, self-regulation, autonomy, agency, task value, interest, engagement, and achievement-related beliefs.	A Single-Item Measure for Assessing STEM Identity Keywords: STEM identity; STEM attitudes; STEM self-efficacy; agency; goals
Language, Literacy, Reading and Bilingual Education	38	Work on language learning, reading, literacy, writing, bilingualism, dyslexia, vocabulary, speech perception, tones, cognates, and language-related assessment.	Development and validation of a rapid and precise online sentence reading efficiency assessment Keywords: dyslexia; reading fluency assessment; screening tools; reading fluency and comprehension; progress monitoring; psychometrics
Open Science, Research Methods and Meta-Research	32	Meta-level communities concerned with open science, preregistration, bibliometrics, systematic reviews, taxonomies, methodological reviews, secondary data, and research infrastructure.	Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database Keywords: ICT; mathematics education; bibliometric analysis; Scopus
Informal Learning, Outreach and Science Communication	18	Research on learning beyond formal classrooms, including museums, outreach, science communication, public engagement, affinity spaces, social media publics, family/community learning, and outreach programs.	Characterizing an online, science-based affinity space using topic modelling, diversity indices, and social network analysis Keywords: Machine learning; science communication; science outreach; InfoMaps; Netlytic; NodeXL

Analytic interpretation

The strongest visible cross-cutting communities are assessment/measurement, STEM and disciplinary education, higher education/student success, digital learning, teacher education, AI/learning analytics, and equity/inclusion. These communities overlap substantially. For example, a study on automated essay scoring belongs simultaneously to AI/learning analytics, assessment, language/writing, and digital learning. Likewise, a study on STEM identity may bridge STEM education, motivation/identity, assessment, and equity.

This overlap is substantively meaningful: the literature selection does not represent isolated research fields, but a network of communities around educational improvement, measurement, digital transformation, equity, and disciplinary learning.

Top recurring keyword signals

higher education (19), covid-19 (10), stem education (10), motivation (8), assessment (7), virtual reality (7), inclusive education (7), artificial intelligence (6), education (6), learning analytics (6), systematic review (6), children (5), stress (4), intervention (4), science communication (4), self-efficacy (4), psychometrics (4), undergraduate education (4), learning (4), teacher education (4), special education (4), adolescents (4), generative ai (4), online education (4), academic integrity (4), survey (4), program evaluation (3), personality (3), sentiment analysis (3), socioeconomic status (3), curriculum (3), natural language processing (3), diversity (3), machine learning (3), bioinformatics (3)

Coding caveats

Because the coding uses only titles and keywords, it should be treated as a transparent thematic map rather than a full-text content analysis. Some communities may be undercounted when a title or keyword list does not explicitly name the relevant field. Conversely, some articles receive multiple codes because broad concepts such as assessment, digital learning, or equity often appear alongside more specific communities.